Propulsion and machinery arrangement

SDO/PROP/CODAG-009 Rev A

RESTRICTED

3.1 Plant configuration

The combined diesel and gas arrangement pairs two cruise diesel engines with a single boost gas turbine through a cross-connect gearbox. Cruise diesels drive both shafts at speeds up to 18 knots; the gas turbine is clutched in for sprint operation above that. The cross-connect gearbox is the single most space-critical item in the machinery arrangement and its footprint shall be fixed before the main transverse bulkhead positions are frozen.

4.5 Blade loading

Five-bladed controllable pitch propellers shall be adopted to reduce blade rate excitation into the hull. Cavitation inception speed shall be not less than 15 knots at the design draught to preserve the acoustic advantage during transit. Blade area ratio shall be selected so that mean thrust loading does not exceed 130 kN per square metre of expanded blade area at continuous service rating.

2.2 Alignment calculation

Shaft alignment shall be calculated for the hot running condition with the hull in the full load afloat state. No bearing shall be unloaded in any operating condition, and the reaction on the aftmost sterntube bearing shall remain between 60 and 110 percent of its nominal design load.